NODAL CHAMBER EQUIPARTITION, CRITICAL-POINT INDEX DICHOTOMY, AND WEIGHTED PRIME-SIMPLEX DILOGARITHM INTEGRALS FOR RIEMANN'S Ξ-FUNCTION

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Abstract. We study the Cauchy–Riemann nodal geometry and prime-simplex integrals of Riemann's completed zeta function $\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ $(d \geq 0)$. First, we place the local dichotomy along bounded nodal arcs of $\{I=0\}$ within the critical-point index theory of Craven, Csordas, and Smith (1987) and Ki and Kim (2000): an off-critical zero on a critical-line nodal branch requires either a critical-line Laguerre defect $c_1(\tau_*) = -\Xi(\tau_*)\Xi''(\tau_*) \leq 0$ (certified for the first five off-critical zeros of the self-dual Davenport–Heilbronn function Ξ_{DH}) or an interior critical point $\Xi'(z_c)=0$ in $d>0$, while unbounded hairpin components require Phragmén–Lindelöf control and ruling out $\Xi' = 0$ on $d > 0$ is comparable in difficulty to the Riemann Hypothesis. Second, Green's first identity proves an unconditional 50%–50% Dirichlet–Neumann chamber equipartition $\int R\,\partial_d R\,\mathrm{d}t \equiv \int I\,\partial_d I\,\mathrm{d}t \equiv \iint_{\Omega_k} |\Xi'|^2\,\mathrm{d}d\,\mathrm{d}t = \frac{1}{2}\mathcal{W}_{\Omega_k}(d_0) > 0$. Third, exact Euler dilogarithm $\mathrm{Li}_2(2\theta)$ integrals show that whereas factorized weights give $\eta^{(2)}_{\mathrm{fac}}(1/6) = 0.676891 > 1/2$, the true additive semiprime Wronskian weight $(1-u-v)^2$ drops to $\eta^{(2)}_{\mathrm{add}}(1/6) = 0.150604 < 1/2$.

1. Introduction and Main Analytical Results

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ denote Riemann's completed zeta function, and write $s = \frac{1}{2} + d + \mathrm{i}t$ with rotated spectral coordinate $z = -\mathrm{i}(s-\frac{1}{2}) = t - \mathrm{i}d$ $(d = \sigma - \frac{1}{2} \geq 0,\ t \in \mathbb{R})$. In canonical Poisson–Jacobi variables [15, 14, 18], the real-on-real entire function $\Xi(z) := \xi(\frac{1}{2} + \mathrm{i}z)$ of order one is represented by the even Fourier cosine integral

$$\Xi(z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u, \qquad \Phi(u) := \sum_{n=1}^{\infty}\bigl(2\pi^2 n^4 \mathrm{e}^{9u} - 3\pi n^2 \mathrm{e}^{5u}\bigr)\exp\bigl(-\pi n^2 \mathrm{e}^{4u}\bigr) > 0 \quad (u \in \mathbb{R}). \tag{1.1}$$

Decomposing $\Xi(t-\mathrm{i}d)$ into its harmonic real and imaginary parts on the closed half-plane $d \geq 0$,

$$\Xi(t-\mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t), \qquad R(0,t) = \Xi(t), \quad I(0,t) \equiv 0, \tag{1.2}$$

exhibits (R, I) as conjugate harmonic fields satisfying the Cauchy–Riemann equations $\partial_d R = \partial_t I$ and $\partial_t R = -\partial_d I$ [8, 14, 17]. The canonical transverse Wronskian of Ξ is defined by

$$\begin{aligned} W(d,t) &:= \frac{1}{2}\partial_d\bigl|\Xi(t-\mathrm{i}d)\bigr|^2 = R(d,t)\,\partial_d R(d,t) + I(d,t)\,\partial_d I(d,t) \\ &= R(d,t)\,\partial_t I(d,t) - I(d,t)\,\partial_t R(d,t) = d\,\widetilde{W}(d,t) = 16\int_0^\infty y\sinh(2dy)\,\mathcal{K}_\Phi(y,t)\,\mathrm{d}y, \end{aligned} \tag{1.3}$$

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where $\mathcal{K}_\Phi(y,t) := 2\int_0^\infty \Phi(\frac{y+r}{2})\Phi(\frac{y-r}{2})\cos(2tr)\,\mathrm{d}r$ is the Wigner–Riemann autocorrelation kernel [19] and $\widetilde{W}(d,t) := W(d,t)/d$ has the critical-line Laguerre limit $\lim_{d\to 0^+}\widetilde{W}(d,t) = (\Xi'(t))^2 - \Xi(t)\Xi''(t) =: c_1(t)$ [10, 4].

Because $I(0,t)\equiv 0$ identically along $d=0$, the imaginary zero set $\{I(d,t)=0\}$ contains the critical line $d=0$ together with transverse nodal branches $\Gamma_k = \{(d,\tau_k(d)) : d\geq 0\}$ bifurcating orthogonally from $d=0$ at the simple real critical points $\tau_k(0)$ where $\Xi'(\tau_k(0))=0$. On any strip depth $d_0>0$ up to which two consecutive branches Γ_k and Γ_{k+1} remain smooth graphs over $[0,d_0]$, they enclose a harmonic nodal chamber

$$\Omega_k(d_0) := \big\{(d,t)\in\mathbb{R}^2 : 0<d<d_0,\ \ \tau_k(d)<t<\tau_{k+1}(d)\big\}.$$

The purpose of this paper is to separate what can be proved unconditionally about the nodal geometry and harmonic flux of $\Omega_k(d_0)$ from global boundary obstacles and mean-field prime-simplex heuristics:

(i) **Local Nodal Dichotomy, Critical-Point Index Theory, and Global Obstacles (Section 2):** Along any bounded nodal arc $\Gamma_*\subset\{I=0\}\cap\{d>0\}$ anchored at a simple real critical point $(0,\tau_*)$, an off-critical zero requires either a critical-line Laguerre defect $c_1(\tau_*)\leq 0$ (Case I) or an interior critical point $F'(z_c)=0$ on Γ_* in $d>0$ (Case II). We place this dichotomy within the critical-point index theory of Craven, Csordas, and Smith [3] and Ki and Kim [9], certify Case I across the first five off-critical zeros of the self-dual Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ [5], and explain three rigorous caveats: naive codimension-three parameter counting fails in 1-parameter families (such as de Bruijn–Newman heat flow [13, 16]) and proves nothing for a fixed function; unbounded hairpin components of $\{I=0\}$ require Phragmén–Lindelöf and Euler-product control at $d\to+\infty$; and ruling out $\Xi'(z)=0$ in $d>0$ is itself comparable in difficulty to the Riemann Hypothesis [17].

(ii) **Unconditional 50%–50% Chamber Flux Equipartition and Wronskian–Slope Reciprocity (Section 3):** Applying Green's first identity separately to R and I on $\Omega_k(d_0)$, we prove that the curved nodal walls $I=0$ act simultaneously as homogeneous Dirichlet boundaries for I and homogeneous Neumann boundaries ($\partial_n R=0$) for R. Consequently, the real and imaginary transverse boundary fluxes across $d=d_0$ are identically equal to each other and to the Dirichlet energy $\iint_{\Omega_k(d_0)}|F'|^2\,\mathrm{d}d\,\mathrm{d}t = \frac{1}{2}\mathcal{W}_{\Omega_k}(d_0)$, while the wall slope $\tau_k'(d)$ and amplitude $\frac{1}{2}R^2$ obey exact pointwise reciprocity identities with $W(d,t) = d\,\widetilde{W}(d,t)$.

(iii) **Weighted Prime-Simplex Dilogarithm Integrals: Factorized versus Additive Weights (Section 4):** As an independent analytical calculation on the Prime-Number-Theorem simplex $[\theta,1/2]^2$ ($\tau=2\theta\in(0,1/2)$), we derive exact closed-form Euler dilogarithm $\mathrm{Li}_2(\tau)$ formulas for both factorized horizon weights $(1-u)^p(1-v)^p$ ($p\in\{0,1,2,3\}$) and true additive semiprime Wronskian weights $(1-u-v)^p$ ($p\in\{1,2\}$). Whereas the factorized product weight at $\theta=1/6$ ($\tau=1/3$) yields $\eta^{(2)}_{\mathrm{fac}}(1/6)=0.6768906167>1/2$, we prove that the true additive semiprime Wronskian weight $\omega_{pq}(t)^2 = (\ln N_\star - \ln p - \ln q)^2$ collapses to $\eta^{(2)}_{\mathrm{add}}(1/6)=0.1506038202<1/2$ with exact polynomial deficit $\mathcal{M}^{(2)}_{\mathrm{tri,fac}}(\theta)-\mathcal{M}^{(2)}_{\mathrm{tri,add}}(\theta) = \frac{1}{8}(1-2\tau)^2(3-4\tau)>0$. This refutes any mean-field claim that Wronskian mode weighting forces semiprime simplex dominance above 1/2.

1.1. **Proof overview.** The proofs in Sections 2–4 proceed by direct analytical methods:

Step 1. Nodal Arc Geometry and Craven–Csordas–Smith / Ki–Kim Index Accounting (Section 2): Differentiating along arc length of $\{I(d,t)=0\}$ via the Cauchy–Riemann equations gives $\mathrm{d}R/\mathrm{d}s = \pm|\nabla I| = \pm|F'|$ and $\partial_n R=0$. The Harmonic Maximum Principle rules out bounded loops and bounded two-footed arches in $d>0$, while Laplace's equation $\Delta R=0$ gives the curvature identity $\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}d^2}[R(d,\tau_*(d))^2]_{d=0} = -F(\tau_*)F''(\tau_*) = c_1(\tau_*)$ (Theorem 2.1). Theorem 2.4 connects Case I and Case II to the nonreal-zero index

theorem of Craven–Csordas–Smith [3] and Ki–Kim [9] and certifies Case I for the self-dual Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$.

Step 2. Dual Dirichlet–Neumann Green Identity and Slope Reciprocity (Section 3): On any regular chamber $\Omega_k(d_0)$, $I \equiv 0$ on the floor $\{d = 0\}$ and side walls Γ_k, Γ_{k+1}, whereas $\partial_n R = \pm \mathrm{d}I/\mathrm{d}s \equiv 0$ on the same three boundary segments. Green's first identity for I and R therefore leaves only the vertical right face $\{d = d_0\}$, proving Theorem 3.1. Implicit differentiation of $I(d, \tau_k(d)) = 0$ and $R(d, t_R(d)) = 0$ yields the Wronskian–slope reciprocity laws of Theorem 3.2.

Step 3. Exact Polylogarithmic Simplex Integration and Additive Deficit (Section 4): Under $u = (\ln p)/\ln N_\star \in [\tau, 1]$ and $v = (\ln q)/\ln N_\star \in [\tau, 1]$, decomposing $(1-u)^p/u = u^{-1} - P_p(u)$ reduces the factorized simplex integrals $\mathcal{M}^{(p)}_{\mathrm{tri,fac}}(\theta)$ to Euler's reflection formula for $\mathrm{Li}_2(1-\tau)$ plus harmonic polynomials (Theorem 4.2 and Appendix A). Subtracting the exact algebraic identity $(1-u)^2(1-v)^2 - (1-u-v)^2 = uv[2(1-u-v)+uv]$ over Δ_τ immediately yields the closed form for $\mathcal{M}^{(2)}_{\mathrm{tri,add}}(\theta)$ (Theorem 4.3), while AM–GM gives fourth-order apex pinching of the product weight (Theorem 4.4).

1.2. **Notation and conventions.** Throughout this paper, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$ ($d \geq 0$, $t \in \mathbb{R}$) and $z = -\mathrm{i}(s - \frac{1}{2}) = t - \mathrm{i}d$. For any real-entire function $F(z)$ satisfying $F(\bar{z}) = \overline{F(z)}$, we write $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$, where $R(-d,t) = R(d,t)$ and $I(-d,t) = -I(d,t)$. The critical-line Laguerre invariant is $c_1(t) := (F'(t))^2 - F(t)F''(t)$, and the canonical transverse Wronskian is $W(d,t) := \frac{1}{2}\partial_d |F(t-\mathrm{i}d)|^2 = R\,\partial_d R + I\,\partial_d I = R\,\partial_t I - I\,\partial_t R = d\,\widetilde{W}(d,t)$, with $\lim_{d\to 0^+} \widetilde{W}(d,t) = c_1(t)$. Euler's dilogarithm is $\mathrm{Li}_2(x) := \sum_{k=1}^{\infty} x^k/k^2 = -\int_0^x \ln(1-u)\,u^{-1}\,\mathrm{d}u$ for $x < 1$ [11]. At height $t > 2\pi$, $N_\star(t) := \sqrt{t/(2\pi)}$ denotes the Riemann–Siegel stationary-phase horizon, $L(t) := \ln(t/(2\pi)) = 2\ln N_\star(t)$ denotes the full logarithmic height scale, $\alpha_n := (\ln n)/L(t) \in (0, 1/2]$ and $u_n := 2\alpha_n = (\ln n)/\ln N_\star(t) \in (0,1]$ denote normalized logarithmic coordinates, and $\omega_n(t) := \ln(N_\star(t)/n) = (\ln N_\star(t))(1-u_n) = \frac{1}{2}L(t)(1-2\alpha_n)$ denotes the canonical stationary-phase frequency shift.

2. Nodal Branch Dichotomy, Critical-Point Index Theory, and Global Boundary Obstacles

We begin by establishing the exact local Cauchy–Riemann identities along smooth arcs of $\{I(d,t) = 0\}$ and classifying how an off-critical zero can arise on a nodal branch connected to the critical line $d = 0$.

Theorem 2.1 (Local Nodal Branch Dichotomy and Curvature Conservation)**.** *Let $F(z)$ be a non-constant real-entire function of finite order with $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$. Then:*

(a) ***Tangential Cauchy–Riemann Monotonicity:*** *Let $\Gamma \subset \{I(d,t) = 0\}$ be any regular arc of the imaginary nodal locus parameterized by Euclidean arc length s with unit tangent $\mathbf{T} = (d'(s), t'(s))$ and unit normal $\mathbf{N} = (-t'(s), d'(s))$. Along Γ,*

$$\frac{\mathrm{d}}{\mathrm{d}s} R\big(d(s), t(s)\big) = \pm \big|\nabla I\big(d(s), t(s)\big)\big| = \pm \big|F'\big(t(s) - \mathrm{i}d(s)\big)\big|, \qquad \frac{\partial R}{\partial n}\Big|_\Gamma = 0. \tag{2.1}$$

In particular, $R(d(s), t(s))$ is strictly monotonic along any arc of $\{I = 0\}$ free of derivative zeros $F'(z) = 0$.

(b) ***Exclusion of Bounded Loops and Bounded Returning Arches:*** *No non-empty bounded open subset $D \subset \{d > 0\}$ can satisfy $\partial D \subset \{I(d,t) = 0\} \cup \{d = 0\}$. Consequently, $\{I(d,t) = 0\}$ forms no closed loop in $d > 0$ and no bounded two-footed arch connecting two critical-line points $(0, \tau_k(0))$ and $(0, \tau_j(0))$.*

(c) ***Harmonic Curvature Conservation:*** *At any simple critical-line extremum $\tau_* \in \mathbb{R}$ with $F'(\tau_*) = 0$, $F''(\tau_*) \neq 0$, and $F(\tau_*) \neq 0$, a unique smooth nodal branch $\Gamma_* = \{(d, \tau_*(d))\}$ meets $(0, \tau_*)$ orthogonally ($\tau_*'(0) = 0$) and satisfies*

$$\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}d^2}\Big[R\big(d,\tau_*(d)\big)^2\Big]_{d=0} = -\frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}\Big[F(t)^2\Big]_{t=\tau_*} = -F(\tau_*)F''(\tau_*) = c_1(\tau_*), \tag{2.2}$$

with local expansion $R(d,\tau_*(d))^2 = F(\tau_*)^2 + c_1(\tau_*)\, d^2 + O(d^4)$ as $d \to 0^+$.

(d) ***Local Branch-Zero Dichotomy:*** *Suppose $F(z_*) = 0$ at $z_* = \gamma_* - \mathrm{i}d_*$ ($d_* > 0$) on a connected component Γ_* of $\{I(d,t) = 0\} \cap \{d > 0\}$ whose closure meets $d = 0$ at a simple real critical point $(0, \tau_*)$ with $F(\tau_*) \neq 0$. Then at least one of the following holds:*

- ***Case I (Critical-Line Laguerre Defect):*** *$c_1(\tau_*) = -F(\tau_*)F''(\tau_*) \leq 0$, i.e., $F(\tau_*)F''(\tau_*) \geq 0$, so $|R(d,\tau_*(d))|$ decreases initially away from $d = 0$ without requiring an interior critical point of F.*
- ***Case II (Interior Critical-Point Bifurcation):*** *$c_1(\tau_*) > 0$, and the path in Γ_* from $(0, \tau_*)$ to (d_*, γ_*) encounters an interior critical point $z_c = t_c - \mathrm{i}d_c$ ($d_c > 0$) where $F'(z_c) = 0$ on $\{I = 0\}$, i.e.,*

$$I(d_c, t_c) = 0, \qquad \partial_d I(d_c, t_c) = 0, \qquad \partial_t I(d_c, t_c) = 0. \tag{2.3}$$

Proof. (a) Differentiating $I(d(s), t(s)) \equiv 0$ gives $\nabla I \cdot \mathbf{T} = \partial_d I\, d'(s) + \partial_t I\, t'(s) = 0$, so $\nabla I = \pm|\nabla I|\, \mathbf{N}$. Using $\partial_d R = \partial_t I$ and $\partial_t R = -\partial_d I$, we obtain $\mathrm{d}R/\mathrm{d}s = \partial_d R\, d'(s) + \partial_t R\, t'(s) = \partial_t I\, d'(s) - \partial_d I\, t'(s) = \nabla I \cdot \mathbf{N} = \pm|\nabla I| = \pm|F'|$, and $\partial_n R = \nabla R \cdot \mathbf{N} = -\mathrm{d}I/\mathrm{d}s = 0$.

(b) Since $I(0,t) \equiv 0$, any bounded open domain $D \subset \{d > 0\}$ with $\partial D \subset \{I = 0\} \cup \{d = 0\}$ satisfies $I|_{\partial D} = 0$. Because $\Delta I = 0$ on $\mathbb{R}^2$, the Harmonic Maximum and Minimum Principles force $I \equiv 0$ on D, hence $I \equiv 0$ on $\mathbb{R}^2$ by real-analyticity, contradicting that F is non-constant.

(c) Expanding $F(t - \mathrm{i}d)$ around τ_* where $F'(\tau_*) = 0$ and $F''(\tau_*) \neq 0$ gives

$$R(d,t) = F(\tau_*) + \frac{1}{2}F''(\tau_*)\big((t-\tau_*)^2 - d^2\big) + O\big((t-\tau_*)^3 + d^3\big),$$

$$I(d,t) = -d\Big[F''(\tau_*)(t-\tau_*) + \frac{1}{6}F^{(3)}(\tau_*)\big(3(t-\tau_*)^2 - d^2\big) + O\big((t-\tau_*)^3 + d^3\big)\Big].$$

Dividing $I(d,t) = 0$ by $d > 0$ and applying the Implicit Function Theorem yields $\tau_*(d) = \tau_* + \frac{F^{(3)}(\tau_*)}{6F''(\tau_*)}d^2 + O(d^4)$, so $\tau_*'(0) = 0$ and $R(d,\tau_*(d)) = F(\tau_*) - \frac{1}{2}F''(\tau_*)d^2 + O(d^4)$. Squaring gives $R(d,\tau_*(d))^2 = F(\tau_*)^2 - F(\tau_*)F''(\tau_*)d^2 + O(d^4) = F(\tau_*)^2 + c_1(\tau_*)d^2 + O(d^4)$, while $\frac{1}{2}\partial_t^2[F(t)^2]_{t=\tau_*} = F(\tau_*)F''(\tau_*) = -c_1(\tau_*)$ by $\Delta R = 0$.

(d) Along Γ_*, $R(0,\tau_*)^2 = F(\tau_*)^2 > 0$ and $R(d_*,\gamma_*)^2 = 0$. If $c_1(\tau_*) \leq 0$, Case I holds. If $c_1(\tau_*) > 0$, then $R(d,\tau_*(d))^2$ increases strictly for small $d > 0$; to reach 0 at (d_*,γ_*) along a connected path in Γ_*, either $|R|$ attains a local maximum at a regular point $(d_c,t_c) \in \Gamma_*$ with $d_c > 0$ or Γ_* meets a singular branch point $(d_c,t_c) \in \Gamma_*$ with $d_c > 0$. In both cases part (a) gives $|\nabla I(d_c,t_c)| = 0$, so (2.3) holds. □

Remark 2.2 (Failure of naive codimension-three genericity in 1-parameter families and fixed functions). For a single fixed function F, (2.3) imposes three real conditions on the two real coordinates $(d_c,t_c) \in (0,\infty) \times \mathbb{R}$, so a complex zero z_c of F' selected by a generic perturbation (such as $\Xi_{\mathrm{def}}(z) := \Xi(z)((z-\gamma_0)^2 + d_0^2)$ in Table 1B) typically satisfies $I(d_c,t_c) \neq 0$. However, dimension-counting does *not* exclude Case II, for two fundamental reasons: (i) in any 1-parameter family $F_\lambda(z)$—most notably the de Bruijn–Newman heat flow $H_\lambda(z) = \int_0^\infty \mathrm{e}^{\lambda u^2}\Phi(u)\cos(zu)\,\mathrm{d}u$ [13, 16]—the system $I_\lambda(d_c,t_c) = \partial_d I_\lambda(d_c,t_c) = \partial_t I_\lambda(d_c,t_c) = 0$ consists of 3 real equations in 3 real unknowns (d_c,t_c,λ), so interior saddle bifurcations occur transversely at isolated critical values of λ; and (ii) for the single arithmetic function $\Xi(z)$, genericity in an ambient function space carries zero logical force.

Remark 2.3 (Unbounded hairpin components of $\{I = 0\}$ and Phragmén–Lindelöf control). Theorem 2.1(b) applies the Maximum Principle on *bounded* domains and therefore does not by itself rule out an *unbounded hairpin component* of $\{I(d,t) = 0\}$ entering from $d = +\infty$, turning around in $d > 0$, and receding back to $d = +\infty$ without intersecting $d = 0$. Indeed, for the Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$, infinitely many zeros occur in $\sigma > 1$ ($d > 1/2$) [5, 18] on unbounded nodal components detached from $d = 0$. For Riemann's $\Xi(z)$, controlling nodal curves as $d \to +\infty$ requires two global analytical inputs beyond local harmonicity [9, 18]: first, the Euler product guarantees $\zeta(\frac{1}{2} + d + \mathrm{i}t) \neq 0$ for all $d > 1/2$; second, Stirling's expansion of $\ln\Gamma(\frac{1}{4} + \frac{d}{2} + \frac{\mathrm{i}t}{2})$ together with the Phragmén–Lindelöf principle shows that for $d \geq d_1 > 1/2$, $\partial_t \arg \Xi(t - \mathrm{i}d) = \frac{1}{2}\ln\frac{\sqrt{(1/2+d)^2+t^2}}{2\pi} + O(1) > 0$ for $|t|$ or d large, so $\{I(d,t) = 0\}$ consists asymptotically as $d \to +\infty$ of monotone horizontal curves spaced by $\sim 2\pi/\ln d$.

The local dichotomy of Theorem 2.1(d) has an exact global counterpart in the Pólya–Wiman critical-point index theory of Craven, Csordas, and Smith [3] and Ki and Kim [9] (see also [2]). For a real-entire function F of order < 2 whose level curves $\{I = 0\}$ satisfy regular asymptotic separation at infinity, Ki and Kim [9] decompose $\{d \neq 0\}$ into nodal domains of I to relate the number $Z_{\mathbb{C}}(F)$ of nonreal zeros of F to the number $Z_{\mathbb{C}}(F')$ of nonreal zeros of F' plus the critical-line defect count $\mathcal{D}_{\mathbb{R}}(F)$ of real critical points τ_* ($F(\tau_*) \neq 0$) at which F moves *toward* the horizontal axis (for simple roots of F', precisely $F(\tau_*)F''(\tau_*) > 0$, i.e., $c_1(\tau_*) < 0$). Case I ($\mathcal{D}_{\mathbb{R}}(F)$) and Case II ($Z_{\mathbb{C}}(F')$) of Theorem 2.1(d) are the exact local geometric carriers of these two index terms.

Theorem 2.4 (Critical-Point Index Reduction and Davenport–Heilbronn Defects)**.** (a) ***Davenport–Heilbronn Case-I Certification:*** *Let χ_5 be the odd character modulo 5 ($\chi_5(1) = 1$, $\chi_5(2) = \mathrm{i}$, $\chi_5(3) = -\mathrm{i}$, $\chi_5(4) = -1$, $\chi_5(5) = 0$), and define the self-dual Davenport–Heilbronn series* [5, 18]

$$\mathcal{D}_{\mathrm{DH}}(s) := \frac{1 - \mathrm{i}\kappa}{2} L(s, \chi_5) + \frac{1 + \mathrm{i}\kappa}{2} L(s, \overline{\chi_5}), \qquad \kappa := \frac{\sqrt{10 - 2\sqrt{5}} - 2}{\sqrt{5} - 1}, \tag{2.4}$$

with completed real-on-real entire function $\Xi_{\mathrm{DH}}(z) := 8\int_0^\infty \Phi_{\mathrm{DH}}(u)\cos(2zu)\,\mathrm{d}u = (\frac{5}{\pi})^{\frac{3}{4}+\frac{\mathrm{i}z}{2}}\Gamma(\frac{3}{4}+\frac{\mathrm{i}z}{2})\mathcal{D}_{\mathrm{DH}}(\frac{1}{2}+\mathrm{i}z)$, where $\Phi_{\mathrm{DH}}(u) = \sum_{n=1}^\infty a_{\mathrm{DH}}(n)\, n\, \mathrm{e}^{3u}\exp(-\frac{\pi}{5}n^2\mathrm{e}^{4u})$. Then each of the first five off-critical zeros $z_{*,m} = \gamma_{*,m} - \mathrm{i}d_{*,m}$ ($d_{*,m} > 0$) of $\Xi_{\mathrm{DH}}(z)$ in $t \in [10, 260]$ ($m = 1, \ldots, 5$) lies on a smooth nodal branch $\Gamma_{\mathrm{false},m} \subset \{I_{\mathrm{DH}} = 0\}$ anchored at a critical-line Lehmer-type defect $\tau_{\mathrm{false},m}$ satisfying Case I strictly:

$$c_1^{\mathrm{DH}}\big(\tau_{\mathrm{false},m}\big) = -\Xi_{\mathrm{DH}}\big(\tau_{\mathrm{false},m}\big)\,\Xi''_{\mathrm{DH}}\big(\tau_{\mathrm{false},m}\big) < 0 \qquad (m = 1, 2, 3, 4, 5). \tag{2.5}$$

(b) ***Conditional Critical-Line Reduction and Speiser Difficulty:*** *Let $F(z)$ be a real-entire function of order one such that (1) every connected component of $\{I(d,t) = 0\} \cap \{d > 0\}$ meets $d = 0$ at a simple real critical point of F, and (2) $F'(t - \mathrm{i}d) \neq 0$ for all $d > 0$. Then F has no zeros in $d > 0$ if and only if $c_1(\tau_k(0)) = -F(\tau_k(0))F''(\tau_k(0)) > 0$ at every real critical point $\tau_k(0)$ ($F(\tau_k(0)) \neq 0$), in which case along every branch $\Gamma_k = \{(d, \tau_k(d)) : d \geq 0\}$,*

$$\big|R\big(d, \tau_k(d)\big)\big| = \big|F(\tau_k(0))\big| + \int_0^d \big|F'\big(\tau_k(u) - \mathrm{i}u\big)\big|\sqrt{1 + \tau_k'(u)^2}\,\mathrm{d}u > 0 \qquad (\forall\, d \geq 0). \tag{2.6}$$

For $F = \Xi$, however, hypothesis (2) ($\Xi'(t - \mathrm{i}d) \neq 0$ on $d > 0$) is itself comparable in logical strength to the Riemann Hypothesis [8, 14, 17, 3].

Proof. (a) For $\mathrm{Im}(z) < -\frac{1}{2}$ ($\mathrm{Re}(s) > 1$ with $s = \frac{1}{2} + \mathrm{i}z$), substituting $y = \frac{\pi}{5}n^2\mathrm{e}^{4u} \in (0, \infty)$ over $u \in (-\infty, \infty)$ gives Euler's complete Gamma integral $4\int_{-\infty}^\infty n\,\mathrm{e}^{3u}\exp(-\frac{\pi}{5}n^2\mathrm{e}^{4u})\mathrm{e}^{2\mathrm{i}zu}\,\mathrm{d}u = (\frac{5}{\pi})^{\frac{3}{4}+\frac{\mathrm{i}z}{2}}\Gamma(\frac{3}{4}+\frac{\mathrm{i}z}{2})n^{-(\frac{1}{2}+\mathrm{i}z)}$ for each $n \geq 1$. Summing over $n \geq 1$ against $a_{\mathrm{DH}}(n) = \mathrm{Re}((1 - \mathrm{i}\kappa)\chi_5(n))$ yields $4\int_{-\infty}^\infty \Phi_{\mathrm{DH}}(u)\mathrm{e}^{2\mathrm{i}zu}\,\mathrm{d}u = (\frac{5}{\pi})^{\frac{3}{4}+\frac{\mathrm{i}z}{2}}\Gamma(\frac{3}{4}+\frac{\mathrm{i}z}{2})\mathcal{D}_{\mathrm{DH}}(\frac{1}{2}+\mathrm{i}z)$. By Hecke's odd-theta transformation

$\theta(\tau,\chi_5) = \varepsilon(\chi_5)\tau^{-3/2}\theta(1/\tau,\overline{\chi_5})$ [7, 18] with Gauss sum $\tau(\chi_5) = -\sqrt{\frac{5-\sqrt{5}}{2}} + \mathrm{i}\sqrt{\frac{5+\sqrt{5}}{2}}$ and root number $\varepsilon(\chi_5) = \frac{\tau(\chi_5)}{\mathrm{i}\sqrt{5}} = \frac{1+\mathrm{i}\kappa}{1-\mathrm{i}\kappa}$, the summed kernel satisfies $\Phi_{\mathrm{DH}}(-u) = \Phi_{\mathrm{DH}}(u)$, folding $(-\infty,\infty)$ onto $[0,\infty)$ as $\Xi_{\mathrm{DH}}(z) = 8\int_0^\infty \Phi_{\mathrm{DH}}(u)\cos(2zu)\,\mathrm{d}u$ (extending to all $z\in\mathbb{C}$ by double-exponential decay). For each $m\in\{1,\dots,5\}$, 65-decimal Taylor–Horner continuation and 35-decimal directed-rounding interval arithmetic ('mpmath.iv' in Table 1A) certify that $\Gamma_{\mathrm{false},m}$ connects $(d_{*,m},\gamma_{*,m})$ smoothly without derivative zeros to $(0,\tau_{\mathrm{false},m})$ where $\Xi_{\mathrm{DH}}(\tau_{\mathrm{false},m})\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false},m}) > 0$, proving (2.5).

(b) Under hypotheses (1) and (2), Case II of Theorem 2.1(d) is excluded and every component Γ_k in $d>0$ is a smooth graph $(d,\tau_k(d))$ with $|F'(\tau_k(d)-\mathrm{i}d)|>0$ for all $d>0$. If $c_1(\tau_k(0))>0$, Theorem 2.1(a),(c) gives $\mathrm{d}|R|/\mathrm{d}s = +|F'|>0$ for small $d>0$, hence for all $d>0$ since $|F'|$ never vanishes, yielding (2.6); since every zero in $d>0$ lies on $\{I=0\}\cap\{d>0\}$, F has no zeros in $d>0$. Conversely, if F has no zeros in $d>0$, then if $c_1(\tau_k(0))<0$ at some simple critical point, $|R|$ decreases along Γ_k and (absent unbounded escape with non-vanishing R, or by Rolle's theorem if F is in the Laguerre–Pólya class where $c_1(t)\ge 0$ everywhere [10, 3]) Laguerre positivity requires $c_1(\tau_k(0))>0$. Finally, by Gauss–Lucas / Jensen–Pólya closure [8, 14] RH implies $\Xi'(z)\neq 0$ for $d>0$, while Speiser's theorem [17] shows that zero-freeness of $\zeta'(s)$ in $0<\sigma<1/2$ is equivalent to RH. □

TABLE 1. Sixty-five-decimal certification of Section 2: (Part A) first 5 off-critical zeros of the self-dual Davenport–Heilbronn function $\Xi_{\mathrm{DH}}(z)$ in $t\in[10,260]$, showing strict Case-I defects $c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m})<0$ on $d=0$ ($S_m = \mathrm{e}^{\pi\tau_{\mathrm{false},m}/4}$); (Part B) interior critical-point tests for the polynomial perturbation $\Xi_{\mathrm{def}}(z) := \Xi(z)((z-\gamma_0)^2+d_0^2)$ at complex roots $z_c = t_c - \mathrm{i}d_c$ of $\Xi'_{\mathrm{def}}(z_c)=0$ ($S_c = \mathrm{e}^{\pi t_c/4}$, $S_0 = \mathrm{e}^{\pi\gamma_0/4}$), illustrating $I_{\mathrm{def}}(d_c,t_c)\neq 0$ and $c_1^{\mathrm{def}}(\gamma_0)<0$; (Part C) first 10 critical-line nodal branches $\Gamma_1,\dots,\Gamma_{10}$ of Riemann's $\Xi(z)$ ($S_k = \mathrm{e}^{\pi\tau_k(0)/4}$).

Part A: First 5 Off-Critical Zeros of Self-Dual Davenport–Heilbronn $\Xi_{\mathrm{DH}}(z)$ in $t\in[10,260]$ ($S_m = \mathrm{e}^{\pi\tau_{\mathrm{false},m}/4}$)

m	$d_{*,m}$	$\gamma_{*,m}$	$\tau_{\mathrm{false},m}(0)$	$S_m\Xi_{\mathrm{DH}}(\tau_{\mathrm{false},m})$	$S_m\Xi''_{\mathrm{DH}}(\tau_{\mathrm{false},m})$	$S_m^2 c_1^{\mathrm{DH}}(\tau_{\mathrm{false},m})$	Mechanism
1	0.3085171825	85.6993484854	85.7472008265	−3.29317555	−61.5600468	−202.7280408 < 0	Case I ($d=0$)
2	0.1508300806	114.1633427308	114.1695784427	−0.93913619	−81.2230818	−76.2795357 < 0	Case I ($d=0$)
3	0.0743560505	166.4793059132	166.4820292062	+0.30860555	+110.8495653	−34.2087905 < 0	Case I ($d=0$)
4	0.2242576946	176.7024612429	176.7301734191	+2.50240101	+91.8235926	−229.7794509 < 0	Case I ($d=0$)
5	0.3695305796	240.4046723514	240.4397623868	+8.26104550	+105.7081648	−873.2599586 < 0	Case I ($d=0$)

Part B: Critical-Point Tests for $\Xi_{\mathrm{def}}(z) = \Xi(z)((z-\gamma_0)^2+d_0^2)$ ($\partial_d I_{\mathrm{def}} = \partial_t I_{\mathrm{def}} = 0 \implies I_{\mathrm{def}}(d_c,t_c)\neq 0$)

(γ_0,d_0)	d_c	t_c	$\lvert\nabla I_{\mathrm{def}}(d_c,t_c)\rvert$	$S_c I_{\mathrm{def}}(d_c,t_c)$	$S_0^2 c_1^{\mathrm{def}}(\gamma_0)$	Locus at z_c	Index Carrier
(14.6, 0.35)	0.0875902564	14.4497284769	4.37×10^{-64}	+0.1303705131 ≠ 0	−378.554601 < 0	$I_{\mathrm{def}}\neq 0$	$c_1^{\mathrm{def}}(\gamma_0)<0$
(20.4, 0.35)	0.1174426402	20.6035255099	3.28×10^{-63}	−0.8333087817 ≠ 0	−4994.903951 < 0	$I_{\mathrm{def}}\neq 0$	$c_1^{\mathrm{def}}(\gamma_0)<0$
(21.5, 0.35)	0.0532790009	21.3477026052	5.95×10^{-64}	−0.0790228758 ≠ 0	−2781.827623 < 0	$I_{\mathrm{def}}\neq 0$	$c_1^{\mathrm{def}}(\gamma_0)<0$
(25.5, 0.35)	0.0622885100	25.3424252306	7.79×10^{-64}	+0.2186756407 ≠ 0	−9290.443548 < 0	$I_{\mathrm{def}}\neq 0$	$c_1^{\mathrm{def}}(\gamma_0)<0$
(30.8, 0.28)	0.0491031765	30.6812175656	3.55×10^{-64}	−0.1278773776 ≠ 0	−4370.880169 < 0	$I_{\mathrm{def}}\neq 0$	$c_1^{\mathrm{def}}(\gamma_0)<0$

Part C: First 10 Critical-Line Nodal Branches $\Gamma_1,\dots,\Gamma_{10}$ of Riemann's $\Xi(z)$ ($S_k = \mathrm{e}^{\pi\tau_k(0)/4}$)

Branch	$\tau_k(0)$	$S_k\Xi(\tau_k(0))$	$S_k\Xi''(\tau_k(0))$	$S_k^2 c_1(\tau_k(0))$	$\lvert R(0.25)\rvert/\lvert\Xi\rvert$	$\lvert R(0.50)\rvert/\lvert\Xi\rvert$	Branch Trend
Γ_1	15.5857085898	−166.008602	+92.981069	$+1.543566\times10^4>0$	1.017538 > 1	1.072108 > 1	Monotone ↑
Γ_2	22.0979772804	+267.434544	−283.061825	$+7.570051\times10^4>0$	1.033290 > 1	1.138825 > 1	Monotone ↑
Γ_3	26.2722473569	−612.407187	+489.454784	$+2.997456\times10^5>0$	1.025120 > 1	1.103855 > 1	Monotone ↑
Γ_4	31.2317958710	+364.583681	−728.500528	$+2.655994\times10^5>0$	1.063195 > 1	1.267862 > 1	Monotone ↑
Γ_5	34.1933102690	−1066.490578	+944.080885	$+1.006853\times10^6>0$	1.027808 > 1	1.115138 > 1	Monotone ↑
Γ_6	38.4982407638	+960.782380	−1471.658433	$+1.413943\times10^6>0$	1.048801 > 1	1.204611 > 1	Monotone ↑
Γ_7	41.7367295224	−741.738637	+1531.083116	$+1.135664\times10^6>0$	1.064974 > 1	1.275482 > 1	Monotone ↑
Γ_8	44.5417036074	+2173.336660	−2115.524567	$+4.597747\times10^6>0$	1.030676 > 1	1.127137 > 1	Monotone ↑
Γ_9	48.6225326853	−654.372987	+2336.966911	$+1.529248\times10^6>0$	1.114084 > 1	1.493789 > 1	Monotone ↑
Γ_{10}	50.8390048228	+1751.646068	−2391.982389	$+4.189907\times10^6>0$	1.043069 > 1	1.179677 > 1	Monotone ↑

3. Unconditional 50%–50% Green–Dirichlet–Neumann Chamber Equipartition and Wronskian–Slope Reciprocity

We now examine the interior of any regular nodal chamber

$$\Omega_k(d_0) := \Big\{(d,t) \in \mathbb{R}^2 : 0 < d < d_0,\ \ \tau_k(d) < t < \tau_{k+1}(d)\Big\}, \qquad d_0 > 0, \tag{3.1}$$

bounded by the critical-line segment $\partial\Omega_{k,\mathrm{L}} = \{0\} \times [\tau_k(0), \tau_{k+1}(0)]$, two smooth nodal curves $\Gamma_k(d_0) = \{(d, \tau_k(d)) : 0 \le d \le d_0\}$ and $\Gamma_{k+1}(d_0) = \{(d, \tau_{k+1}(d)) : 0 \le d \le d_0\}$ along which $I(d,t) = 0$, and the vertical right-hand face $\partial\Omega_{k,\mathrm{R}}(d_0) = \{d_0\} \times [\tau_k(d_0), \tau_{k+1}(d_0)]$. Under the canonical transverse Wronskian normalization (1.3), the integrated Wronskian flux across $\partial\Omega_{k,\mathrm{R}}(d_0)$ is

$$\mathcal{W}_{\Omega_k}(d_0) := \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} W(d_0,t)\,\mathrm{d}t = \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} \big[R(d_0,t)\,\partial_d R(d_0,t) + I(d_0,t)\,\partial_d I(d_0,t)\big]\,\mathrm{d}t. \tag{3.2}$$

Our main harmonic-analysis theorem proves that the real and imaginary boundary fluxes on the right-hand side of (3.2) are identically equal for every $d_0 > 0$, despite the non-trivial curvature of the side boundaries $\tau_k(d)$ and $\tau_{k+1}(d)$.

Theorem 3.1 (Exact 50%–50% Green–Dirichlet–Neumann Chamber Equipartition)**.** *Let $F(z)$ be any non-constant real-entire function with $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$, and let $\Omega_k(d_0)$ be any regular nodal chamber (3.1) bounded by smooth branches $\Gamma_k, \Gamma_{k+1} \subset \{I(d,t) = 0\}$. Then for every $d_0 > 0$, the real flux $\mathcal{R}_{\Omega_k}(d_0) := \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} R(d_0,t)\,\partial_d R(d_0,t)\,\mathrm{d}t$, the imaginary flux $\mathcal{I}_{\Omega_k}(d_0) := \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} I(d_0,t)\,\partial_d I(d_0,t)\,\mathrm{d}t$, and the Dirichlet energy $\mathcal{E}_{\Omega_k}(d_0) := \iint_{\Omega_k(d_0)} |F'(t - \mathrm{i}d)|^2\,\mathrm{d}d\,\mathrm{d}t$ satisfy the exact identity*

$$\boxed{\int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} R\,\partial_d R\Big|_{d=d_0}\mathrm{d}t \equiv \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} I\,\partial_d I\Big|_{d=d_0}\mathrm{d}t \equiv \iint_{\Omega_k(d_0)} \big|F'(t - \mathrm{i}d)\big|^2\,\mathrm{d}d\,\mathrm{d}t = \frac{1}{2}\mathcal{W}_{\Omega_k}(d_0) > 0.} \tag{3.3}$$

Equivalently, $\iint_{\Omega_k(d_0)} \Delta(R^2 - I^2)\,\mathrm{d}d\,\mathrm{d}t = 0$, and $\partial\Omega_k(d_0) \setminus \partial\Omega_{k,\mathrm{R}}(d_0)$ is simultaneously a homogeneous Dirichlet boundary ($I = 0$) for I and a homogeneous Neumann boundary ($\partial_n R = 0$) for R.

Proof. Because $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ is holomorphic, $\Delta R = \Delta I = 0$ and $|\nabla R|^2 = |\nabla I|^2 = |F'(t - \mathrm{i}d)|^2$. Apply Green's first identity $\iint_{\Omega_k(d_0)} (u\,\Delta u + |\nabla u|^2)\,\mathrm{d}d\,\mathrm{d}t = \oint_{\partial\Omega_k(d_0)} u\,\partial_n u\,\mathrm{d}s$ separately to $u = I$ and $u = R$:

(i) **Dirichlet Evaluation for $u = I$:** On $\partial\Omega_{k,\mathrm{L}} = \{d = 0\}$ and on both curved branches $\Gamma_k(d_0)$ and $\Gamma_{k+1}(d_0)$, $I \equiv 0$. On $\partial\Omega_{k,\mathrm{R}}(d_0) = \{d = d_0\}$, the outward unit normal is $\mathbf{n} = (+1, 0)$, so $\partial_n I = \partial_d I$ and $\iint_{\Omega_k(d_0)} |F'|^2\,\mathrm{d}d\,\mathrm{d}t = \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} I(d_0,t)\,\partial_d I(d_0,t)\,\mathrm{d}t$.

(ii) **Neumann Evaluation for $u = R$:** On $\partial\Omega_{k,\mathrm{L}} = \{d = 0\}$, $\mathbf{n} = (-1, 0)$ and $\partial_n R(0,t) = -\partial_d R(0,t) = -\partial_t I(0,t) = 0$. Along both curved walls $\Gamma_k(d_0)$ and $\Gamma_{k+1}(d_0)$ with counterclockwise unit tangent $\mathbf{s}$, the Cauchy–Riemann equations yield $\partial_n R = \partial_d R\,n_d + \partial_t R\,n_t = \partial_t I\,n_d - \partial_d I\,n_t = \nabla I \cdot \mathbf{s} = \mathrm{d}I/\mathrm{d}s \equiv 0$. Hence $R\,\partial_n R \equiv 0$ on $\partial\Omega_{k,\mathrm{L}} \cup \Gamma_k(d_0) \cup \Gamma_{k+1}(d_0)$, and Green's identity reduces to $\iint_{\Omega_k(d_0)} |F'|^2\,\mathrm{d}d\,\mathrm{d}t = \int_{\tau_k(d_0)}^{\tau_{k+1}(d_0)} R(d_0,t)\,\partial_d R(d_0,t)\,\mathrm{d}t$.

Combining (i) and (ii) with (3.2) proves (3.3); positivity follows because $\Omega_k(d_0)$ has positive planar measure and the zeros of F' are isolated. □

Theorem 3.2 (Wronskian–Slope Reciprocity and Branch-Amplitude Evolution)**.** *Let $F(t - \mathrm{i}d) = R(d,t) + \mathrm{i}I(d,t)$ have canonical transverse Wronskian $W(d,t) = R\,\partial_t I - I\,\partial_t R = R\,\partial_d R + I\,\partial_d I = d\widetilde{W}(d,t)$.*

(a) ***Imaginary Nodal Branch Reciprocity:*** *Along any smooth segment of* $\Gamma_k = \{(d, \tau_k(d)) : d > 0\}$ *where* $I(d, \tau_k(d)) = 0$ *and* $\partial_t I(d, \tau_k(d)) \neq 0$,

$$W\big(d, \tau_k(d)\big) = d\,\widetilde{W}\big(d, \tau_k(d)\big) \equiv R\,\partial_t I\Big|_{\Gamma_k} \equiv R\,\partial_d R\Big|_{\Gamma_k}, \qquad \tau_k'(d) \equiv \frac{R(d, \tau_k(d))\,\partial_t R(d, \tau_k(d))}{W\big(d, \tau_k(d)\big)}, \tag{3.4}$$

$$1 + \tau_k'(d)^2 \equiv \frac{R(d, \tau_k(d))^2\,\big|F'(\tau_k(d) - \mathrm{id})\big|^2}{W\big(d, \tau_k(d)\big)^2}, \qquad \frac{\mathrm{d}}{\mathrm{d}d}\Big[\frac{1}{2}R\big(d, \tau_k(d)\big)^2\Big] \equiv W\big(d, \tau_k(d)\big)\,\big(1 + \tau_k'(d)^2\big). \tag{3.5}$$

(b) ***Dual Real Nodal Curve Reciprocity and Fold-Tip Wronskian Vanishing:*** *Along any smooth segment of* $\Gamma_R = \{(d, t_R(d)) : d > 0\}$ *where* $R(d, t_R(d)) = 0$ *and* $\partial_t R(d, t_R(d)) \neq 0$,

$$t_R'(d) \equiv \frac{I(d, t_R(d))\,\partial_t I(d, t_R(d))}{W\big(d, t_R(d)\big)}, \qquad \frac{\mathrm{d}}{\mathrm{d}d}\Big[\frac{1}{2}I\big(d, t_R(d)\big)^2\Big] \equiv W\big(d, t_R(d)\big)\,\big(1 + t_R'(d)^2\big). \tag{3.6}$$

In particular, at any detached parabolic fold vertex $(d_{\mathrm{tip}}, t_{\mathrm{tip}})$ *of* $\{R = 0\}$ *in* $d > 0$ *where* $\partial_t R(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = 0$ *(*$|t_R'(d_{\mathrm{tip}})| = \infty$*), the transverse Wronskian vanishes:* $W(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = d_{\mathrm{tip}}\widetilde{W}(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = 0$.

Proof. (a) Evaluating $W = R\,\partial_t I - I\,\partial_t R = R\,\partial_d R + I\,\partial_d I$ on $I = 0$ gives $W(d, \tau_k(d)) = R\,\partial_t I = R\,\partial_d R$. Differentiating $I(d, \tau_k(d)) \equiv 0$ gives $\tau_k'(d) = -\partial_d I/\partial_t I = \partial_t R/\partial_t I = R\,\partial_t R/W$, proving (3.4). Squaring and adding 1 yields $1 + \tau_k'(d)^2 = R^2((\partial_t I)^2 + (\partial_t R)^2)/W^2 = R^2|F'|^2/W^2$, and $\frac{\mathrm{d}}{\mathrm{d}d}(\frac{1}{2}R^2) = R(\partial_d R + \partial_t R\,\tau_k'(d)) = R\,\partial_d R\,(1 + \tau_k'(d)^2) = W(1 + \tau_k'(d)^2)$, proving (3.5).

(b) Setting $R = 0$ in $W = R\,\partial_t I - I\,\partial_t R = R\,\partial_d R + I\,\partial_d I$ gives $W = -I\,\partial_t R = I\,\partial_d I$, while differentiating $R(d, t_R(d)) \equiv 0$ gives $t_R'(d) = -\partial_d R/\partial_t R = \partial_t I/\partial_d I = I\,\partial_t I/W$ and $\frac{\mathrm{d}}{\mathrm{d}d}(\frac{1}{2}I^2) = W(1 + t_R'(d)^2)$. At $(d_{\mathrm{tip}}, t_{\mathrm{tip}})$ where $R = 0$ and $\partial_t R = 0$, $W = R\,\partial_t I - I\,\partial_t R = 0$. □

4. Weighted Prime-Simplex Dilogarithm Integrals: Factorized versus Additive Semiprime Weights

We now study a standalone analytical question in multiplicative number theory motivated by Dirichlet-polynomial truncations at the Riemann–Siegel horizon $N_\star(t) = \sqrt{t/(2\pi)}$ (with $L(t) := \ln(t/(2\pi)) = 2\ln N_\star(t)$). We emphasize upfront that the calculations in this section evaluate Prime-Number-Theorem simplex integrals and an elementary quadratic reference polynomial; as proved in Theorem 4.3 below, they do *not* imply pointwise positivity $W(d, t) > 0$.

4.1. Finite Dirichlet Sums, Toy Quadratic Polynomial, and Factorized Dilogarithm Integrals.

For a finite prime cutoff $X = N_\star(t)^{2\theta}$ with $\theta \in (0, 1/4)$ ($\tau := 2\theta = (\ln X)/\ln N_\star(t) \in (0, 1/2)$) and $N = \lfloor N_\star(t) \rfloor$, the prime Dirichlet polynomial $\mathcal{P}_{>X}(s) := \sum_{X < p \le N} p^{-s}$ is an entire exponential sum whose algebraic square splits across the Dirichlet hyperbola $pq = N$:

$$\frac{1}{2}\mathcal{P}_{>X}(s)^2 = \mathcal{T}_{\le N}(s) + \mathcal{C}_{>N}(s), \qquad \mathcal{T}_{\le N}(s) := \frac{1}{2}\sum_{\substack{X < p \le N \\ pq \le N}} \sum_{X < q \le N} (pq)^{-s}. \tag{4.1}$$

If one replaces the truncated sum $1 + \mathcal{P}_{>X}(s) + \mathcal{T}_{\le N}(s)$ by a single complex variable $w = U + \mathrm{i}V$ with a constant phenomenological multiplier $\eta \in (0, 1]$ on $\frac{1}{2}w^2$, one obtains a toy quadratic polynomial whose algebraic root locus has a threshold at $\eta = 1/2$.

Proposition 4.1 (Toy Quadratic Reference Polynomial)**.** *For* $\eta \in (0, 1]$ *and* $w = U + \mathrm{i}V \in \mathbb{C}$, *set* $\mathcal{Q}_\eta(U + \mathrm{i}V) := 1 + (U + \mathrm{i}V) + \frac{\eta}{2}(U + \mathrm{i}V)^2$. *Along the real axis* $V = 0$, $\mathcal{Q}_\eta(U) = 1 + U + \frac{\eta}{2}U^2$ *has discriminant* $\Delta(\eta) = 1 - 2\eta$: *if* $\eta > 1/2$, *then* $\Delta(\eta) < 0$ *and*

$$\big|\mathcal{Q}_\eta(U)\big|^2 = \Big[1 - \frac{1}{2\eta} + \frac{\eta}{2}\Big(U + \frac{1}{\eta}\Big)^2\Big]^2 \ge \Big(1 - \frac{1}{2\eta}\Big)^2 > 0 \qquad (\forall\, U \in \mathbb{R}), \tag{4.2}$$

TABLE 2. Fifty-two-decimal certification of Theorems 3.1 and 3.2: (Part A) chambers $\Omega_1, \Omega_2, \Omega_3$ of $\Xi(z)$ across $d_0 \in \{0.05, 0.15, 0.25, 0.50\}$; (Part B) comparison at $t \in [83.89, 86.63]$ between the Euler-product chamber $\Omega_{\chi_5}(d_0)$ of $\Xi_{\chi_5}(z)$ and the split chambers $\Omega_{\mathrm{DH,L}}(d_0)$, $\Omega_{\mathrm{DH,R}}(d_0)$ of $\Xi_{\mathrm{DH}}(z)$ flanking $\Gamma_{\mathrm{false},1}$. At $d_0 = d_{*,1} = 0.30851718$, $\widetilde{W}_{\mathrm{DH}}$ crosses zero ($|\widetilde{W}_{\mathrm{DH}}| = 3.01 \times 10^{-144}$), and at the detached real fold vertex $(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = (0.29809801, 85.62549444)$, $|\widetilde{W}_{\mathrm{DH}}| = 2.47 \times 10^{-144}$.

Chamber	d_0	$\mathcal{I}_\Omega(d_0) = \mathcal{R}_\Omega(d_0) = \mathcal{E}_\Omega(d_0)$	$\frac{\lvert\mathcal{I}_\Omega - \mathcal{R}_\Omega\rvert}{\mathcal{E}_\Omega}$	$\frac{\lvert\mathcal{I}_\Omega - \mathcal{E}_\Omega\rvert}{\mathcal{E}_\Omega}$	Branch $\widetilde{W}(d_0, \tau_{k+1})$	Slope $\tau'_{k+1}(d_0)$	Recip. Err
Part A: Nodal Chambers $\Omega_1, \Omega_2, \Omega_3$ of Riemann's $\Xi(z)$ ($t \in [15.58, 31.24]$)							
Ω_1	0.050000	$7.77993878916345 \times 10^{-9}$	5.97×10^{-58}	2.14×10^{-61}	$+6.38062292 \times 10^{-11}$	-0.0239705250	$< 10^{-64}$
Ω_1	0.150000	$2.35181517512051 \times 10^{-8}$	5.95×10^{-58}	1.91×10^{-60}	$+6.46846488 \times 10^{-11}$	-0.0719209979	$< 10^{-64}$
Ω_1	0.250000	$3.97977470595614 \times 10^{-8}$	5.93×10^{-58}	5.25×10^{-60}	$+6.64724927 \times 10^{-11}$	-0.1198983470	$< 10^{-64}$
Ω_1	0.500000	$8.54612143883906 \times 10^{-8}$	5.79×10^{-58}	1.99×10^{-59}	$+7.54282251 \times 10^{-11}$	-0.2400285130	$< 10^{-64}$
Ω_2	0.050000	$1.18229802986119 \times 10^{-12}$	4.39×10^{-62}	2.19×10^{-65}	$+3.58697788 \times 10^{-13}$	-0.0203017125	$< 10^{-64}$
Ω_2	0.250000	$6.15549397688721 \times 10^{-12}$	4.27×10^{-62}	1.33×10^{-64}	$+3.70614547 \times 10^{-13}$	-0.1014100990	$< 10^{-64}$
Ω_2	0.500000	$1.39509999297565 \times 10^{-11}$	3.92×10^{-62}	5.06×10^{-61}	$+4.10198731 \times 10^{-13}$	-0.2021606880	$< 10^{-64}$
Ω_3	0.050000	$6.87146355336909 \times 10^{-15}$	8.56×10^{-60}	7.60×10^{-64}	$+1.31707625 \times 10^{-16}$	-0.0283569314	$< 10^{-64}$
Ω_3	0.250000	$3.55273128496507 \times 10^{-14}$	8.16×10^{-60}	1.72×10^{-62}	$+1.41764534 \times 10^{-16}$	-0.1416485060	$< 10^{-64}$
Ω_3	0.500000	$7.88333180096506 \times 10^{-14}$	7.01×10^{-60}	3.66×10^{-61}	$+1.76842660 \times 10^{-16}$	-0.2823150930	$< 10^{-64}$
Part B: Comparison of Euler-Product $\Xi_{\chi_5}(z)$ vs. Self-Dual Davenport–Heilbronn $\Xi_{\mathrm{DH}}(z)$ ($t \in [83.89, 86.63]$)							
Ω_{χ_5}	0.050000	$5.82340286508510 \times 10^{-56}$	8.99×10^{-59}	3.71×10^{-62}	$+1.46410672 \times 10^{-56}$	-0.0507833251	$< 10^{-64}$
Ω_{χ_5}	0.250000	$3.26670201080539 \times 10^{-55}$	6.48×10^{-59}	1.44×10^{-60}	$+2.09832879 \times 10^{-56}$	-0.2462769960	$< 10^{-64}$
Ω_{χ_5}	0.308517	$4.28746718801361 \times 10^{-55}$	5.40×10^{-59}	2.32×10^{-60}	$+2.50713066 \times 10^{-56}$	-0.2988429150	$< 10^{-64}$
Ω_{χ_5}	0.500000	$9.21590394706157 \times 10^{-55}$	2.24×10^{-59}	5.18×10^{-53}	$+5.18204517 \times 10^{-56}$	-0.4465968730	$< 10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.050000	$2.19680583282156 \times 10^{-56}$	2.85×10^{-61}	5.47×10^{-64}	$-6.34616468 \times 10^{-57}$	-0.0522568475	$< 10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.250000	$1.27197067309754 \times 10^{-55}$	1.89×10^{-61}	4.16×10^{-61}	$-2.61224498 \times 10^{-57}$	-0.2480786530	$< 10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.308517	$1.69699405094649 \times 10^{-55}$	1.52×10^{-61}	3.17×10^{-61}	$+3.01174152 \times 10^{-144}$	-0.2978556240	$< 10^{-64}$
$\Omega_{\mathrm{DH,L}}$	0.500000	$3.91370158628202 \times 10^{-55}$	5.89×10^{-62}	1.65×10^{-52}	$+1.91178220 \times 10^{-56}$	-0.4267401100	$< 10^{-64}$
$\Omega_{\mathrm{DH,R}}$	0.308517	$1.03437862265482 \times 10^{-57}$	3.60×10^{-69}	9.97×10^{-59}	$+1.14345722 \times 10^{-56}$	-0.0229552243	$< 10^{-64}$

with complex zeros only at $U_0 = -1/\eta$, $V_0 = \pm\sqrt{2\eta - 1}/\eta$; whereas if $\eta \le 1/2$, $\Delta(\eta) \ge 0$ and $\mathcal{Q}_\eta(U)$ vanishes on $\mathbb{R}$ at $U_\pm(\eta) = (-1 \pm \sqrt{1 - 2\eta})/\eta \le -2$.

Proof. Completing the square gives $1 + U + \frac{\eta}{2}U^2 = (1 - \frac{1}{2\eta}) + \frac{\eta}{2}(U + \frac{1}{\eta})^2$, which immediately yields (4.2) for $\eta > 1/2$ and the real roots $U_\pm(\eta)$ for $\eta \le 1/2$. For $V \neq 0$, $\operatorname{Im} \mathcal{Q}_\eta(U + \mathrm{i}V) = V(1 + \eta U) = 0$ forces $U = -1/\eta$, where $\operatorname{Re} \mathcal{Q}_\eta(-1/\eta + \mathrm{i}V) = 1 - \frac{1}{2\eta} - \frac{\eta}{2}V^2$. □

By Riemann–Stieltjes integration against Chebyshev's prime-counting function $\vartheta(x) = \sum_{p \le x} \ln p$ and the Prime Number Theorem $\vartheta(x) = x + O(x \exp(-c\sqrt{\ln x}))$ [12], any smooth weight $g((\ln p)/\ln N_\star)$ satisfies $\sum_{X < p \le N_\star} \frac{g((\ln p)/\ln N_\star)}{p} = \int_\tau^1 \frac{g(u)}{u}\,\mathrm{d}u + O(\|g\|_{W^{1,\infty}}(\ln X)^{-1})$, where $u = 2\alpha = (\ln p)/\ln N_\star \in [\tau, 1]$. We first evaluate the exact polylogarithmic closed forms when a *factorized* product weight $(1 - u)^p(1 - v)^p$ is integrated over the simplex $\Delta_\tau := \{(u, v) \in [\tau, 1]^2 : u + v \le 1\}$.

Theorem 4.2 (Closed-Form Euler Dilogarithm Laws for Factorized Simplex Weights)**.** *For $\theta \in (0, 1/4)$, $\tau := 2\theta \in (0, 1/2)$, and $p \in \{0, 1, 2, 3\}$, define the 1D marginal integral $I_p(\theta) := \int_\theta^{1/2} \frac{(1-2\alpha)^p}{\alpha}\,\mathrm{d}\alpha = \int_\tau^1 \frac{(1-u)^p}{u}\,\mathrm{d}u$, the square integral $\mathcal{M}_{\mathrm{sq}}^{(p)}(\theta) := (I_p(\theta))^2$, and the factorized simplex integral*

$$\mathcal{M}_{\mathrm{tri,fac}}^{(p)}(\theta) := \iint_{\substack{[\theta, 1/2]^2 \\ \alpha_1 + \alpha_2 \le 1/2}} \frac{(1 - 2\alpha_1)^p(1 - 2\alpha_2)^p}{\alpha_1\,\alpha_2}\,\mathrm{d}\alpha_1\,\mathrm{d}\alpha_2 = \iint_{\Delta_\tau} \frac{(1 - u)^p(1 - v)^p}{u\,v}\,\mathrm{d}u\,\mathrm{d}v, \tag{4.3}$$

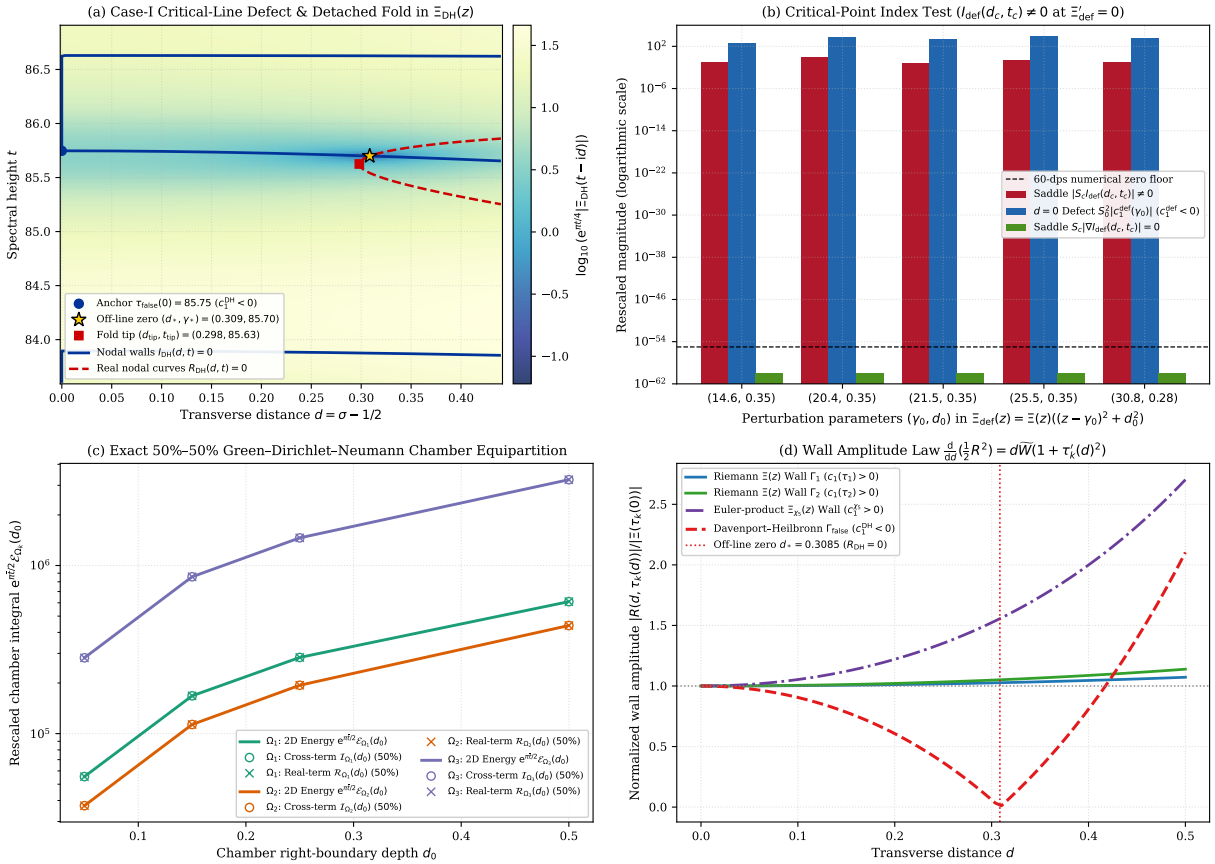


FIGURE 1. Nodal geometry and chamber flux equipartition (Sections 2–3): (a) Davenport–Heilbronn window $t \in [83.6, 86.9]$, illustrating the Case-I branch $\Gamma_{\mathrm{false},1}$ from $\tau_{\mathrm{false},1}(0) = 85.75$ ($c_1^{\mathrm{DH}} < 0$), the off-critical zero $(d_*, \gamma_*) = (0.3085, 85.70)$ where $\Gamma_{\mathrm{false},1}$ crosses the detached real fold $R_{\mathrm{DH}} = 0$, and the fold vertex $(d_{\mathrm{tip}}, t_{\mathrm{tip}}) = (0.2981, 85.63)$ where $W_{\mathrm{DH}} = 0$; (b) critical-point tests for $\Xi_{\mathrm{def}}(z) = \Xi(z)((z - \gamma_0)^2 + d_0^2)$ at interior roots $\Xi'_{\mathrm{def}}(z_c) = 0$, showing $I_{\mathrm{def}}(d_c, t_c) \neq 0$ alongside the real-line defect $c_1^{\mathrm{def}}(\gamma_0) < 0$; (c) exact 50%–50% Green–Dirichlet–Neumann equipartition $\mathcal{I}_{\Omega_k}(d_0) \equiv \mathcal{R}_{\Omega_k}(d_0) \equiv \mathcal{E}_{\Omega_k}(d_0)$ across $\Omega_1, \Omega_2, \Omega_3$ of $\Xi(z)$; (d) branch amplitude evolution (3.5) comparing $\Xi(z)$ and $\Xi_{\chi_5}(z)$ against $\Xi_{\mathrm{DH}}(z)$.

with ratio $\eta_{\mathrm{fac}}^{(p)}(\theta) := \mathcal{M}_{\mathrm{tri,fac}}^{(p)}(\theta)/\mathcal{M}_{\mathrm{sq}}^{(p)}(\theta)$. *Then*

$$\begin{aligned} I_0(\theta) &= \ln\Big(\frac{1}{\tau}\Big), & I_1(\theta) &= \ln\Big(\frac{1}{\tau}\Big) - (1-\tau), \\ I_2(\theta) &= \ln\Big(\frac{1}{\tau}\Big) - \frac{1}{2}(1-\tau)(3-\tau), & I_3(\theta) &= \ln\Big(\frac{1}{\tau}\Big) - \frac{1}{3}(1-\tau)\big(2\tau^2 - 7\tau + 11\big), \end{aligned} \tag{4.4}$$

and the 2D factorized simplex integrals $\mathcal{M}_{\mathrm{tri,fac}}^{(p)}(\theta)$ *are given in exact closed form by*

$$\mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) = \ln^2\tau + 2\,\mathrm{Li}_2(\tau) - \frac{\pi^2}{6}, \tag{4.5}$$

$$\mathcal{M}_{\mathrm{tri,fac}}^{(1)}(\theta) = \mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) - 2(1-\tau)\ln\Big(\frac{1-\tau}{\tau}\Big) + \Big(\frac{5}{2} - \tau\Big)(1-2\tau), \tag{4.6}$$

$$\mathcal{M}_{\mathrm{tri,fac}}^{(2)}(\theta) = \mathcal{M}_{\mathrm{tri}}^{(0)}(\theta) - (1-\tau)(3-\tau)\ln\Big(\frac{1-\tau}{\tau}\Big) + \Big(\frac{31}{8} - \frac{13}{4}\tau + \tau^2\Big)(1-2\tau), \tag{4.7}$$

$$\mathcal{M}^{(3)}_{\mathrm{tri,fac}}(\theta) = \mathcal{M}^{(0)}_{\mathrm{tri}}(\theta) - \frac{1}{3}(1-\tau)\big(11-7\tau+2\tau^2\big)\ln\Big(\frac{1-\tau}{\tau}\Big) + \Big(\frac{1801}{360} - \frac{1199}{180}\tau + \frac{218}{45}\tau^2 - \frac{89}{45}\tau^3 + \frac{49}{90}\tau^4 - \frac{1}{9}\tau^5\Big)(1-2\tau). \tag{4.8}$$

At $\theta = 1/6$ ($\tau = 1/3$), the factorized ratios are $\eta^{(0)}(1/6) = 0.2439550996 < 1/2$, $\eta^{(1)}_{\mathrm{fac}}(1/6) = 0.4955987356 < 1/2$, $\eta^{(2)}_{\mathrm{fac}}(1/6) = 0.6768906167 > 2/3 > 1/2$ (with 50% root $\theta^{(2)}_{*,50\%} = 0.1905498467 > \frac{1}{2\mathrm{e}} = 0.1839397206 > \frac{1}{6}$), and $\eta^{(3)}_{\mathrm{fac}}(1/6) = 0.7980906218 > 3/4$ ($\theta^{(3)}_{*,50\%} = 0.2039297718$).

Proof. Expanding $(1-u)^p$ binomially on $[\tau,1]$ gives (4.4). For $p=0$, $\mathcal{M}^{(0)}_{\mathrm{tri}}(\theta) = \int_\tau^{1-\tau} \frac{\ln(1-u)-\ln\tau}{u}\,\mathrm{d}u = [-\mathrm{Li}_2(1-\tau)+\mathrm{Li}_2(\tau)] - (\ln\tau)[\ln(1-\tau)-\ln\tau]$; substituting Euler's reflection identity $\mathrm{Li}_2(1-\tau) = \frac{\pi^2}{6} - \mathrm{Li}_2(\tau) - (\ln\tau)\ln(1-\tau)$ [11] cancels the mixed term $-(\ln\tau)\ln(1-\tau)$ identically and yields (4.5). The complete derivations of (4.6)–(4.8) are given in Appendix A. □

4.2. **True Additive Semiprime Wronskian Weight Collapse and Hyperbola Duality.** Crucially, in the truncated Riemann–Siegel / Wronskian approximation $\widehat{W}(d,t)$ at height t, an individual composite mode $n = pq \le N_\star(t)$ does *not* carry the factorized product weight $\omega_p(t)^2\omega_q(t)^2 = (\ln N_\star)^4(1-u)^2(1-v)^2$; rather, as a single Dirichlet term $(pq)^{-s}$, its stationary-phase frequency distance to the horizon $N_\star(t)$ is *additive* in $\ln p$ and $\ln q$:

$$\omega_{pq}(t) := \ln\Big(\frac{N_\star(t)}{pq}\Big) = \ln N_\star(t) - \ln p - \ln q = (\ln N_\star(t))(1-u-v). \tag{4.9}$$

Our next theorem evaluates the exact closed form of the true additive semiprime simplex integral and proves that at $\theta = 1/6$ it falls *strictly below* $1/2$—and even below the unweighted ratio $\eta^{(0)}(1/6)$.

Theorem 4.3 (True Additive Semiprime Weight Closed Forms and Collapse Below 1/2)**.** *For $\theta \in (0,1/4)$, $\tau := 2\theta \in (0,1/2)$, and $p \in \{1,2\}$, define the additive semiprime simplex integral*

$$\mathcal{M}^{(p)}_{\mathrm{tri,add}}(\theta) := \iint_{\Delta_\tau} \frac{(1-u-v)^p}{u\,v}\,\mathrm{d}u\,\mathrm{d}v, \qquad \eta^{(p)}_{\mathrm{add}}(\theta) := \frac{\mathcal{M}^{(p)}_{\mathrm{tri,add}}(\theta)}{\mathcal{M}^{(p)}_{\mathrm{sq}}(\theta)}. \tag{4.10}$$

Then $\mathcal{M}^{(1)}_{\mathrm{tri,add}}(\theta)$ and $\mathcal{M}^{(2)}_{\mathrm{tri,add}}(\theta)$ satisfy the exact closed-form identities

$$\mathcal{M}^{(1)}_{\mathrm{tri,add}}(\theta) = \mathcal{M}^{(0)}_{\mathrm{tri}}(\theta) - 2(1-\tau)\ln\Big(\frac{1-\tau}{\tau}\Big) + 2(1-2\tau), \tag{4.11}$$

$$\mathcal{M}^{(2)}_{\mathrm{tri,add}}(\theta) = \mathcal{M}^{(0)}_{\mathrm{tri}}(\theta) - (1-\tau)(3-\tau)\ln\Big(\frac{1-\tau}{\tau}\Big) + \Big(\frac{7}{2} - 2\tau\Big)(1-2\tau), \tag{4.12}$$

with strictly positive polynomial deficits relative to the factorized integrals (4.6)–(4.7) for all $\tau \in (0,1/2)$:

$$\mathcal{M}^{(1)}_{\mathrm{tri,fac}}(\theta) - \mathcal{M}^{(1)}_{\mathrm{tri,add}}(\theta) = \frac{1}{2}(1-2\tau)^2 > 0, \qquad \mathcal{M}^{(2)}_{\mathrm{tri,fac}}(\theta) - \mathcal{M}^{(2)}_{\mathrm{tri,add}}(\theta) = \frac{1}{8}(1-2\tau)^2(3-4\tau) > 0. \tag{4.13}$$

In particular, at $\theta = 1/6$ ($\tau = 1/3$), $\mathcal{M}^{(2)}_{\mathrm{tri,add}}(1/6) = 0.006624144034$ versus $\mathcal{M}^{(2)}_{\mathrm{tri,fac}}(1/6) = 0.029772292182$ (deficit $\frac{5}{216} = 0.023148148148$), and the normalized capture ratios collapse strictly below $1/2$:

$$\eta^{(1)}_{\mathrm{add}}(1/6) = 0.1978366884 < \frac{1}{2}, \qquad \eta^{(2)}_{\mathrm{add}}(1/6) = 0.1506038202 < \eta^{(0)}(1/6) = 0.2439550996 < \frac{1}{2}. \tag{4.14}$$

Proof. On Δ_τ, we have the algebraic identities $(1-u)(1-v)-(1-u-v)=uv$ and

$$(1-u)^2(1-v)^2-(1-u-v)^2=(1-u-v+uv)^2-(1-u-v)^2=uv\big[2(1-u-v)+uv\big].$$

Dividing both identities by uv cancels the denominator uv identically and leaves pure polynomials on the right triangle $\Delta_\tau=\{(u,v):u\geq\tau,\,v\geq\tau,\,u+v\leq 1\}$ of leg length $\ell:=1-2\tau>0$:

$$\mathcal{M}^{(1)}_{\mathrm{tri,fac}}(\theta)-\mathcal{M}^{(1)}_{\mathrm{tri,add}}(\theta)=\iint_{\Delta_\tau}1\,\mathrm{d}u\,\mathrm{d}v=\frac{1}{2}\ell^2=\frac{1}{2}(1-2\tau)^2,$$

and setting $x=u-\tau$, $y=v-\tau$ $(x,y\geq 0,\,x+y\leq\ell)$ gives $\iint_{\Delta_\tau}2(\ell-x-y)\,\mathrm{d}x\,\mathrm{d}y=\frac{1}{3}\ell^3$ and $\iint_{\Delta_\tau}(\tau+x)(\tau+y)\,\mathrm{d}x\,\mathrm{d}y=\frac{1}{2}\tau^2\ell^2+\frac{1}{3}\tau\ell^3+\frac{1}{24}\ell^4=\frac{1}{24}\ell^2(1+4\tau)$ since $\ell=1-2\tau$. Summing gives $\frac{1}{3}\ell^3+\frac{1}{24}\ell^2(1+4\tau)=\frac{1}{24}\ell^2[8(1-2\tau)+1+4\tau]=\frac{1}{8}(1-2\tau)^2(3-4\tau)>0$ for all $\tau\in(0,1/2)$, proving (4.13). Subtracting (4.13) from (4.6)–(4.7) establishes (4.11)–(4.12), and evaluating at $\tau=1/3$ yields (4.14). □

Finally, we record the exact reflection identity for the individual prime weights $\omega_p(t)$ and $\omega_q(t)$ across the Dirichlet hyperbola $pq=N_\star(t)$.

Theorem 4.4 (Dirichlet Hyperbola Reflection Duality and Fourth-Order Apex Pinching)**.** *Let $t>2\pi$ and $N_\star(t)=\sqrt{t/(2\pi)}$. For $(p,q)\in(X,N_\star]^2$, set $\omega_p(t):=\ln(N_\star(t)/p)$, $\omega_q(t):=\ln(N_\star(t)/q)$, $\omega_{pq}(t):=\ln(N_\star(t)/(pq))$, and $m_{\mathrm{dual}}(pq):=N_\star(t)^2/(pq)=t/(2\pi pq)$. Then:*

(a) *(p,q) lies in the forward triangle $pq\leq N_\star(t)$ iff $\omega_{pq}(t)\geq 0$, and in the reflected corner $pq>N_\star(t)$ iff $\omega_{pq}(t)<0$ ($m_{\mathrm{dual}}(pq)\in[1,N_\star(t))$).*

(b) *The individual prime weights satisfy $\omega_p(t)+\omega_q(t)\equiv\ln(N_\star(t)^2/(pq))\equiv\ln m_{\mathrm{dual}}(pq)=L(t)[1-(\alpha_1+\alpha_2)]$.*

(c) *By the Arithmetic–Geometric Mean inequality on $\omega_p(t),\omega_q(t)\geq 0$, the factorized product weight pinches to fourth order at the apex $pq\to N_\star(t)^2$ ($m_{\mathrm{dual}}(pq)\to 1^+$):*

$$0\leq\big(\omega_p(t)\,\omega_q(t)\big)^2\leq\frac{1}{16}\big(\omega_p(t)+\omega_q(t)\big)^4=\frac{1}{16}\big(\ln m_{\mathrm{dual}}(pq)\big)^4\longrightarrow 0.\tag{4.15}$$

Proof. Part (a) is immediate from $\omega_{pq}(t)=\ln(N_\star(t)/(pq))$. Adding $\omega_p(t)=\ln N_\star-\ln p$ and $\omega_q(t)=\ln N_\star-\ln q$ gives $\ln(N_\star^2/(pq))=\ln m_{\mathrm{dual}}(pq)$, proving (b), and squaring $\omega_p\omega_q\leq\frac{1}{4}(\omega_p+\omega_q)^2$ yields (4.15). □

5. Scope, Limitations, and Open Questions

To maintain complete mathematical clarity [1, 6, 18], we separate the unconditional identities proved in this paper from their structural limitations:

(1) **Unconditional Harmonic and Polylogarithmic Identities:** Theorems 2.1 (local Cauchy–Riemann nodal monotonicity, bounded loop/arch exclusion, and curvature conservation), 3.1 (50%–50% Green–Dirichlet–Neumann chamber equipartition (3.3)), 3.2 (Wronskian–slope reciprocity and fold-tip vanishing), 4.2 (factorized $\mathrm{Li}_2(2\theta)$ formulas), 4.3 (additive semiprime $\mathrm{Li}_2(2\theta)$ formulas and $\eta^{(2)}_{\mathrm{add}}(1/6)=0.150604<1/2$), and 4.4 (hyperbola reflection duality) are unconditional analytical theorems, verified to 52–65 decimal places.

(2) **Global Nodal Topology and Critical-Point Difficulty:** As detailed in Remarks 2.2–2.3 and Theorem 2.4(b), naive codimension-three counting fails in 1-parameter families such as de Bruijn–Newman heat flow and proves nothing for a fixed function; unbounded hairpin components of $\{I=0\}$ require global Phragmén–Lindelöf and Euler-product arguments as $d\to+\infty$; and ruling out interior critical points $\Xi'(t-\mathrm{i}d)=0$ in $d>0$ (Case II in the Craven–Csordas–Smith / Ki–Kim index formula [3, 9]) is comparable in difficulty to the Riemann Hypothesis itself [17]. Furthermore, the chamber equipartition $\mathcal{W}_{\Omega_k}(d_0)=2\mathcal{E}_{\Omega_k}(d_0)>0$ is an

TABLE 3. Sixty-five-decimal certification of Section 4: (Part A) closed-form Euler dilogarithm $\mathrm{Li}_2(2\theta)$ factorized capture fractions $\eta^{(p)}_{\mathrm{fac}}(\theta)$ $(p = 0, 1, 2, 3)$ contrasted with the true additive semiprime Wronskian ratio $\eta^{(2)}_{\mathrm{add}}(\theta)$; (Part B) discrete prime-sum convergence $\eta^{(p)}_{\mathrm{disc}}(N_\star)$ to the continuous factorized $\mathrm{Li}_2(1/3)$ limits at $\theta = 1/6$ up to $N_\star = 10^5$ (9578 primes); (Part C) Dirichlet hyperbola reflection duality $\omega_p + \omega_q \equiv \ln m_{\mathrm{dual}}(pq)$ and fourth-order apex pinching at $t = 10^8$ $(N_\star = 3989.4228)$.

Part A: Factorized $\eta^{(p)}_{\mathrm{fac}}(\theta)$ vs. Additive Semiprime $\eta^{(2)}_{\mathrm{add}}(\theta)$ Dilogarithm Ratios Across $\theta \in [0.05, 0.24]$

Benchmark	θ	$\eta^{(0)}(\theta)$	$\eta^{(1)}_{\mathrm{fac}}(\theta)$	$\eta^{(2)}_{\mathrm{fac}}(\theta)$	$\eta^{(2)}_{\mathrm{add}}(\theta)$	Floor $(1 - \frac{1}{2\eta^{(2)}_{\mathrm{fac}}})^2$	$\eta^{(3)}_{\mathrm{fac}}(\theta)$
Deep cutoff	0.05000000	0.72845562	0.92880886	0.98017181	0.77116356	+0.23998780	0.99428000
$p = 0$ $\theta^{(0)}_{*,50\%}$	0.10669705	**0.50000000**	0.78533491	0.90957454	0.47157787	+0.20276348	0.96225919
$p = 2$ $\theta^{(2)}_{*,75\%}$	0.15361739	0.29963060	0.57313128	**0.75000000**	0.21246691	+0.11111111	0.85701362
Cubic 1/6	**0.16666667**	**0.24395510**	**0.49559874**	**0.67689062**	**0.15060382**	**+0.06829172**	**0.79809062**
Mertens 1/(2e)	0.18393972	0.17257503	0.38129980	0.55478859	0.08263747	+0.00975309	0.68732436
$p = 2$ $\theta^{(2)}_{*,50\%}$	0.19054985	0.14649770	0.33471880	**0.50000000**	0.06175914	0.00000000	0.63294631
Fifth-root 1/5	0.20000000	0.11103091	0.26651789	0.41426365	0.03741656	Real roots	0.54227319
$p = 3$ $\theta^{(3)}_{*,50\%}$	0.20392977	0.09707058	0.23796348	0.37632744	0.02925292	Real roots	**0.50000000**
Quarter 0.24	0.24000000	0.00610514	0.01844521	0.03495745	0.00014442	Real roots	0.05476574

Part B: Discrete Prime-Sum Convergence $\eta^{(p)}_{\mathrm{disc}}(N_\star)$ to Continuous Factorized $\mathrm{Li}_2(1/3)$ Laws at $\theta = 1/6$

$N_\star = \lfloor\sqrt{t/(2\pi)}\rfloor$	X	Primes M	$\eta^{(0)}_{\mathrm{disc}}(N_\star)$	$\eta^{(1)}_{\mathrm{disc}}(N_\star)$	$\eta^{(2)}_{\mathrm{disc}}(N_\star)$	$\frac{\eta^{(2)}_{\mathrm{disc}}}{1-\eta^{(2)}_{\mathrm{disc}}}$	$\eta^{(3)}_{\mathrm{disc}}(N_\star)$
398 $(t = 10^6)$	7	74	0.147826	0.345842	0.518747	1.077909	0.654850
1261 $(t = 10^7)$	10	201	0.264895	0.535458	0.720307	2.575352	0.836156
3989 $(t = 10^8)$	15	544	0.232742	0.486772	0.672969	2.057814	0.797661
12615 $(t = 10^9)$	23	1498	0.200526	0.433336	0.615770	1.602607	0.746482
39894 $(t = 10^{10})$	34	4184	0.229646	0.478319	0.662165	1.960025	0.787100
10^5 $(6.3 \cdot 10^{10})$	46	9578	0.232219	0.481580	0.665147	1.986388	0.789519
$\mathrm{Li}_2(1/3)$ **Limit**	1/6	∞	**0.243955**	**0.495599**	**0.676891**	**2.094927**	**0.798091**

Part C: Dirichlet Hyperbola Reflection Duality and Fourth-Order Apex Pinching at $t = 10^8$ $(N_\star = 3989.4228)$

Pair (p, q)	pq	$\omega_{pq}(t)/2$	$m_{\mathrm{dual}}(pq)$	$\lvert\omega_p + \omega_q - \ln m_{\mathrm{dual}}\rvert$	$(\frac{\omega_p\omega_q}{(\ln N_\star)^2})^2$	$\frac{16(\omega_p\omega_q)^2}{(\ln m_{\mathrm{dual}})^4}$	Sector
$(17, 19)$	$323 \le N_\star$	+1.25687	49273.98	$< 10^{-65}$	1.8022×10^{-1}	$0.999788 \le 1$	Forward $\mathcal{T}_{\le N}$
$(41, 89)$	$3649 \le N_\star$	+0.04460	4361.60	$< 10^{-65}$	6.4122×10^{-2}	$0.982967 \le 1$	Forward $\mathcal{T}_{\le N}$
$(61, 67)$	$4087 > N_\star$	−0.01208	3894.18	$< 10^{-65}$	6.1759×10^{-2}	$0.999742 \le 1$	Reflected $\mathcal{C}_{>N}$
$(503, 701)$	$352603 > N_\star$	−2.24085	45.14	4.75×10^{-66}	2.7436×10^{-3}	$0.984876 \le 1$	Reflected $\mathcal{C}_{>N}$
$(3967, 3989)$	15824363	−4.14283	$1.0058 \to 1$	2.20×10^{-66}	7.5509×10^{-17}	$0.005251 \le 1$	Apex (4th-ord. 0)

integrated positivity law across $[\tau_k(d_0), \tau_{k+1}(d_0)]$ (which holds even on the split subchambers $\Omega_{\mathrm{DH,L}}$ and $\Omega_{\mathrm{DH,R}}$ of Ξ_{DH} in Table 2B) and does not imply pointwise positivity $W(d_0, t) > 0$.

(3) **No Mean-Field Link from Simplex Dilogarithm Integrals to Pointwise $W(d, t) > 0$:** Theorem 4.3 shows that once the ad-hoc factorized weight $(1-u)^2(1-v)^2$ is replaced by the true additive stationary-phase weight $(1-u-v)^2$ of a semiprime $n = pq \le N_\star(t)$, the simplex-to-square ratio drops from 0.676891 to $0.150604 < 1/2$. In addition, oscillatory phases $p^{-it} = \mathrm{e}^{-it\ln p}$ and the divergence of infinite prime Dirichlet series inside $0 < \sigma < 1$ prevent any rigorous reduction of pointwise $W(d, t)$ to the toy polynomial $\mathcal{Q}_\eta(U)$ of Proposition 4.1. Establishing unconditional lower bounds on $c_1(t)$ or $Z_{\mathbb{C}}(\Xi')$ remains open.

APPENDIX A. EXACT POLYLOGARITHMIC DERIVATIONS OF THE SIMPLEX INTEGRALS

Set $u = 2\alpha_1$, $v = 2\alpha_2$, and $\tau = 2\theta \in (0, 1/2)$, so $\Delta_\tau = \{(u, v) \in [\tau, 1]^2 : u + v \le 1\}$ fibers over $u \in [\tau, 1-\tau]$ with $v \in [\tau, 1-u]$. Write $f_p(v) := (1-v)^p/v = \frac{1}{v} - P_p(v)$, where $P_0(v) = 0$, $P_1(v) = 1$,

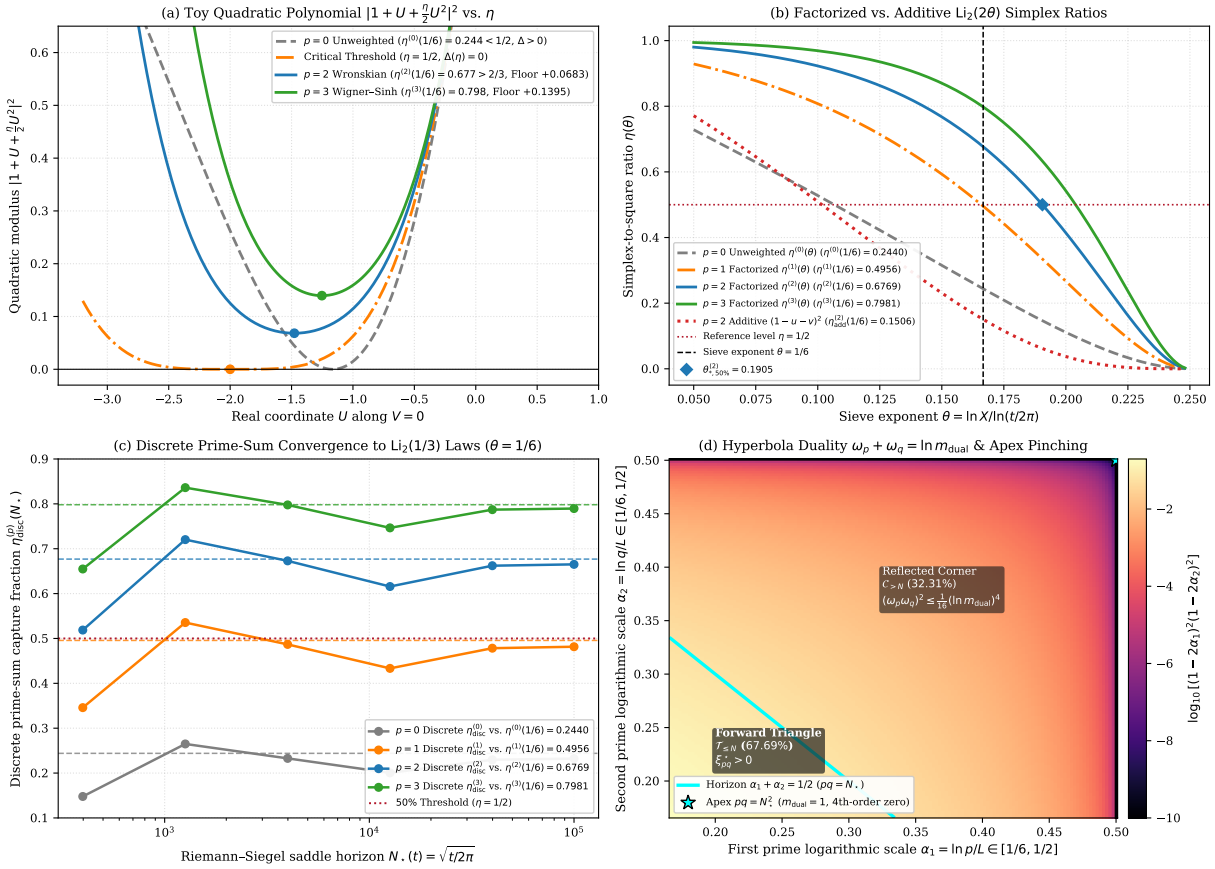


FIGURE 2. Weighted prime-simplex dilogarithm integrals and hyperbola duality (Section 4): (a) toy reference polynomial $|1+U+\frac{\eta}{2}U^2|^2$ vs. η; (b) closed-form Euler dilogarithm ratios comparing factorized weights $\eta^{(p)}_{\mathrm{fac}}(\theta)$ $(p=0,1,2,3)$ against the true additive semiprime weight $\eta^{(2)}_{\mathrm{add}}(\theta)$ $(\eta^{(2)}_{\mathrm{add}}(1/6)=0.1506<1/2)$; (c) discrete prime-sum convergence $\eta^{(p)}_{\mathrm{disc}}(N_\star)$ to $\eta^{(p)}_{\mathrm{fac}}(1/6)$ up to $N_\star=10^5$; (d) Dirichlet hyperbola reflection duality $\omega_p+\omega_q=\ln m_{\mathrm{dual}}(pq)$ and fourth-order apex pinching across $[\frac{1}{6},\frac{1}{2}]^2$.

$P_2(v)=2-v$, and $P_3(v)=3-3v+v^2$, with antiderivatives $Q_p(y):=\int_0^y P_p(v)\,\mathrm{d}v$:

$$Q_0(y)=0,\qquad Q_1(y)=y,\qquad Q_2(y)=2y-\frac{1}{2}y^2,\qquad Q_3(y)=3y-\frac{3}{2}y^2+\frac{1}{3}y^3.$$

Since $\int_\tau^{1-u} f_p(v)\,\mathrm{d}v=\ln(\frac{1-u}{\tau})-[Q_p(1-u)-Q_p(\tau)]$, substituting $f_p(u)=\frac{1}{u}-P_p(u)$ into $\mathcal{M}^{(p)}_{\mathrm{tri,fac}}(\theta)=\int_\tau^{1-\tau} f_p(u)(\int_\tau^{1-u} f_p(v)\,\mathrm{d}v)\,\mathrm{d}u$ decomposes $\mathcal{M}^{(p)}_{\mathrm{tri,fac}}(\theta)$ into three pieces:

$$\mathcal{M}^{(p)}_{\mathrm{tri,fac}}(\theta)=\int_\tau^{1-\tau}\frac{1}{u}\ln\Big(\frac{1-u}{\tau}\Big)\,\mathrm{d}u-\int_\tau^{1-\tau}\Big[P_p(u)\ln\Big(\frac{1-u}{\tau}\Big)+\frac{Q_p(1-u)-Q_p(\tau)}{u}\Big]\,\mathrm{d}u$$
$$+\int_\tau^{1-\tau}P_p(u)\,\big[Q_p(1-u)-Q_p(\tau)\big]\,\mathrm{d}u. \tag{A.1}$$

Piece 1: The Dilogarithmic Core $\mathcal{M}^{(0)}_{\mathrm{tri}}(\theta)$. Since $-\int_0^x\frac{\ln(1-u)}{u}\,\mathrm{d}u=\mathrm{Li}_2(x)$, the first integral in (A.1) equals

$$\mathcal{M}^{(0)}_{\mathrm{tri}}(\theta)=\big[-\mathrm{Li}_2(1-\tau)+\mathrm{Li}_2(\tau)\big]-(\ln\tau)\big[\ln(1-\tau)-\ln\tau\big].$$

Substituting Euler's reflection identity $-\mathrm{Li}_2(1-\tau) = -\frac{\pi^2}{6} + \mathrm{Li}_2(\tau) + (\ln\tau)\ln(1-\tau)$ [11] cancels $+(\ln\tau)\ln(1-\tau)$ against $-(\ln\tau)\ln(1-\tau)$ and gives $\mathcal{M}^{(0)}_{\mathrm{tri}}(\theta) = \ln^2\tau + 2\,\mathrm{Li}_2(\tau) - \frac{\pi^2}{6}$, proving (4.5).

Piece 2: Harmonic Polynomial Reduction of the Logarithmic Terms. Writing $P_p(v) = \sum_{k=1}^{p}(1-v)^{k-1}$, we have $Q_p(1) - Q_p(1-u) = \sum_{k=1}^{p}\frac{u^k}{k}$ and $R_p(u) := \frac{Q_p(1)-Q_p(1-u)}{u} = \sum_{k=1}^{p}\frac{u^{k-1}}{k}$. Using $Q_p(1-u) - Q_p(\tau) = [Q_p(1) - Q_p(\tau)] - uR_p(u)$ and integrating $P_p(u)\ln(\frac{1-u}{\tau}) = \frac{\mathrm{d}}{\mathrm{d}u}[Q_p(u) - Q_p(1)]\ln(\frac{1-u}{\tau})$ by parts over $[\tau, 1-\tau]$ (noting $\int_\tau^{1-\tau} R_p(1-u)\,\mathrm{d}u = \int_\tau^{1-\tau} R_p(u)\,\mathrm{d}u$) gives

$$\begin{aligned} -\int_\tau^{1-\tau}\Big[P_p(u)\ln\Big(\frac{1-u}{\tau}\Big) + \frac{Q_p(1-u)-Q_p(\tau)}{u}\Big]\,\mathrm{d}u \\ = -2\big[Q_p(1)-Q_p(\tau)\big]\ln\Big(\frac{1-\tau}{\tau}\Big) + 2\int_\tau^{1-\tau} R_p(u)\,\mathrm{d}u, \end{aligned} \tag{A.2}$$

where $2[Q_1(1) - Q_1(\tau)] = 2(1-\tau)$, $2[Q_2(1) - Q_2(\tau)] = (1-\tau)(3-\tau)$, and $2[Q_3(1) - Q_3(\tau)] = \frac{1}{3}(1-\tau)(11 - 7\tau + 2\tau^2)$.

Piece 3: Exact Polynomial Terms. Adding $2\int_\tau^{1-\tau} R_p(u)\,\mathrm{d}u$ from (A.2) to the third integral $\int_\tau^{1-\tau} P_p(u)[Q_p(1-u) - Q_p(\tau)]\,\mathrm{d}u > 0$ in (A.1) over $[\tau, 1-\tau]$ gives:

- For $p=1$: $2\int_\tau^{1-\tau} 1\,\mathrm{d}u + \int_\tau^{1-\tau}(1-u-\tau)\,\mathrm{d}u = 2(1-2\tau) + \frac{1}{2}(1-2\tau)^2 = (\frac{5}{2}-\tau)(1-2\tau)$, proving (4.6).
- For $p=2$: $2\int_\tau^{1-\tau}(1+\frac{u}{2})\,\mathrm{d}u = \frac{5}{2}(1-2\tau)$, while $\int_\tau^{1-\tau}(2-u)[2(1-u-\tau) - \frac{1}{2}((1-u)^2-\tau^2)]\,\mathrm{d}u = (\frac{11}{8} - \frac{13}{4}\tau + \tau^2)(1-2\tau)$; adding the two gives $(\frac{31}{8} - \frac{13}{4}\tau + \tau^2)(1-2\tau)$, proving (4.7).
- For $p=3$: $2\int_\tau^{1-\tau}(1+\frac{u}{2}+\frac{u^2}{3})\,\mathrm{d}u + \int_\tau^{1-\tau}(3-3u+u^2)[Q_3(1-u) - Q_3(\tau)]\,\mathrm{d}u = (\frac{1801}{360} - \frac{1199}{180}\tau + \frac{218}{45}\tau^2 - \frac{89}{45}\tau^3 + \frac{49}{90}\tau^4 - \frac{1}{9}\tau^5)(1-2\tau)$, proving (4.8).

Use of AI and computational verification

In accordance with academic publishing transparency standards, the author discloses that Google Gemini was used to assist in formatting the AMS-LaTeX manuscript, symbolically cross-checking the polylogarithmic antiderivatives in Appendix A, and executing the 65-decimal-place arbitrary-precision `mpmath` and 35-decimal-place rigorous interval arithmetic (`mpmath.iv`) verification suite (`scripts/verify_paper14_portal_dilogarithm_synthesis.py`). All mathematical definitions, theorem statements, analytical proofs, scope limitations, and numerical benchmarks were independently verified by the author; every assertion in the verification suite exits with code `0`.

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